stamp) receipt. Table A.9 (row "Type") formalizes this result: in the control group, differences between the specified demographic group and out-group are statistically significant for girls, Hispanic children, and low-income children, and large but imprecise for Black children. These disparities mirror known statistics showing children with these demographic characteristics tend to face heightened risk of child mortality (Currie and Schwandt, 2016) and/or maltreatment (Putnam-Hornstein et al., 2021).

Point estimates suggest that access to the algorithm reduced harm for children in disadvantaged categories, namely Black, Hispanic, female, and low-income children (compare light blue triangles to dark blue triangles in Figure 1). The out-groups (square dots) also experienced improvements in outcomes, but they were smaller in magnitude. Precise estimates are reported in Table A.9, and imply a reduction in disparities in the harm index on the order of 50 to 80 percent (Table A.9, bottom row). Though large, these reductions are not statistically distinguishable from zero.[13]

Our findings are nevertheless informative in a Bayesian perspective. In the absence of prior evidence, suppose a policymaker would have assumed effects on disparities could be of any value. We model this with a uniform prior distribution where algorithm support could reduce disparities by up to 200 percent or increase them by that same amount. Following Brännlund et al. (2024), we then calculate the posterior distribution given the data we observe, assuming normally distributed point estimates with observed variance.[14] We report these posterior distributions with 90 percent credible intervals in Appendix Figure A.8. Our estimates reveal that in most cases we can rule out a worsening of disparities for ethnicity, gender, and socioeconomic status.

4.3 Impact on Racial Disparities in CPS Screen-Ins

Providing algorithmic information may affect disparities in agency surveillance. CPS investigations can sometimes be intrusive and traumatic, and prior work has documented substantial racial disparities – in some cases unwarranted – in CPS contact across the U.S. (Kim et al. 2017, Edwards et al. 2021, Baron et al. 2023). Though a small share of the sample, Black children are nearly twice as likely to be investigated as other children in the sample, even after conditioning on algorithm risk score (Table A.10). In contrast, we find no evidence of screen-in disparities by gender, ethnicity, or socioeconomic status even after controlling

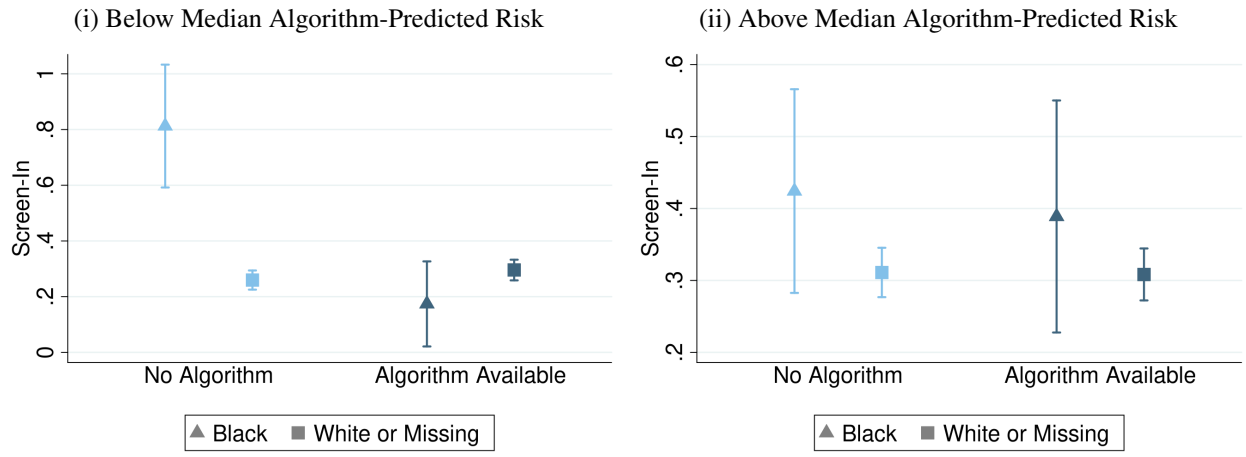
[13]Earlier versions of the paper used Stata's `suest` command, which reported a highly significant joint test $p < 0.01$. However, `suest` does not accommodate different variables temporarily sharing the same name across specifications (i.e., a loop over four disadvantaged groups of interest), leading to an error when compared with hand-calculated stacked regression specifications.

[14]The posterior is a truncated normal distribution over [-200,200]: $p(\theta|\hat{\theta}) = \frac{1}{\Phi(\frac{200-\hat{\theta}}{\sigma})-\Phi(\frac{-200-\hat{\theta}}{\sigma})}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(\hat{\theta}-\theta)^2}{2\sigma^2}\right)$ if $\theta \in [-200, 200]$ and zero otherwise. Φ denotes a normal cumulative distribution function.

for algorithm-predicted risk (Table A.10). We therefore focus our analysis of surveillance disparities on race, specifically Black children relative to children who are not Black (primarily white).

Figure 2 shows that, in the control group, Black children at below-median predicted risk were *60 percentage points* more likely to be screened in than other (primarily white) children of similar risk level. However, when the algorithm was available, below-median-risk Black children were screened in at much lower rates, comparable to rates for other children. Column 2 of Appendix Table A.11 shows that the algorithm tool altogether reduced screen-ins by 13 percentage points for Black children across the entire sample, though point estimates are imprecisely estimated. These findings suggest that access to the algorithm reduced unwarranted racial disproportionality in investigations among low-risk children. Although the trial county has a relatively small Black population, there were still $n = 170$ Black children in the full sample when using CPS-derived race alone, with at least 23 children in each race-by-treatment-status by above or below median risk cell. Regression estimates and robustness to including a permutation test and alternate ways of defining race are reported in Appendix Table A.11. Although the exact magnitude of screen-in disparity reductions – and corresponding power to detect significant effects – depends on the specification, all estimates suggest the algorithm reduced surveillance disparities conditional on risk and, taking the imprecise point estimates at face value, may have canceled out disparities entirely. Though we are statistically underpowered, we do not detect an offsetting increase in the harm index among low-scoring Black children (Appendix Figure A.10i).

Figure 2: Algorithm Tool Reduced Racial Disparities in CPS Surveillance for Low-Risk-Score Children



Notes: These figures present rates of CPS screen-in (investigation) by race and tool availability. Above and below-median predicted risk are defined using the algorithm referral risk score. Figures define race as recorded by Child Protective Services. 90% confidence intervals are reported and randomization controls are included throughout.

These results align with an alternative approach for measuring unwarranted disparities (UDs) that makes no assumptions about whether high or low risk scores merit investigations: comparing harm for screened-in versus screened-out Black and white children, respectively. Under the algorithm, screened-out Black children saw disproportionately greater reductions in harm relative to screened-out white children, with no offsetting change for screened-in children. We show in Section 5 that targeting of investigations (measured by isolating harm for screened-out children) improved more for Black than white children (Figure 3), implying a potential reduction in UDs.

Providing an algorithm risk score could alter stereotypes that workers use to make assessments in the absence of an external benchmark. This is consistent with theoretical work by Bordalo et al. (2016) on how stereotypes form from a representative agent. For low-scoring Black children, workers may have been prompted by the algorithm to change their reference means to that of other low-risk-score children. Discussions with workers underscored their cultural sensitivity and aspiration for greater equality. Future work should examine whether similar reductions in disparities occur in settings with different family demographics or worker beliefs and training.

5 Mechanisms

Providing workers with an algorithm that predicts child risk improved child outcomes and reduced racial disparities. In this section, we link these benefits to improved worker allocation of investigations and rule out a number of alternative mechanisms.

5.1 Improved Worker Allocation of Screen-Ins

The intervention did not increase the total number of investigations ("screen-ins") or any other measurable level of services (Figure A.11). Workers did not increase overall levels of interventions and did not significantly change discussion length, as is sometimes observed due to asymmetric valuation of false positives and false negatives (Almog et al. 2024, Fitzpatrick et al. 2022). In our setting, levels of intervention may have been relatively inelastic given legal, financial, and time constraints.

Notably, workers also did not increase their rate of investigation for children with high algorithm risk scores. Figure A.12 shows that the relationship between screen-in rates and risk scores is essentially flat in the control group and changes little with the introduction of the tool.[15] Although this may appear counterin-

[15] An earlier draft of this paper showed a small-in-magnitude, marginally significant ($p = 0.098$) reduction in the screen-in by